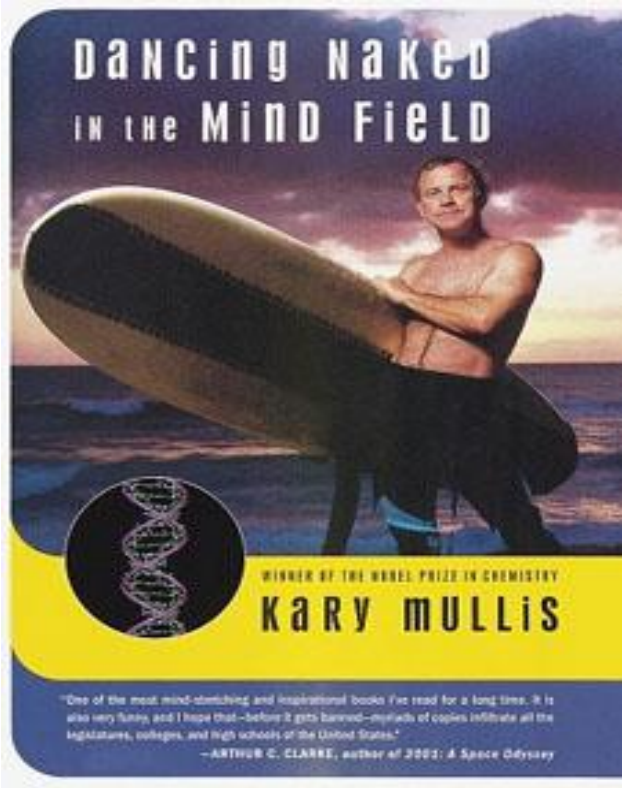
Basic Principles and Components of PCR

DATING IS HEALTHY FOR NAME AND FAME



Unusual Origin of PCR, Mullis KB, Scientific American 1990,56

Development/Invention of PCR Technique

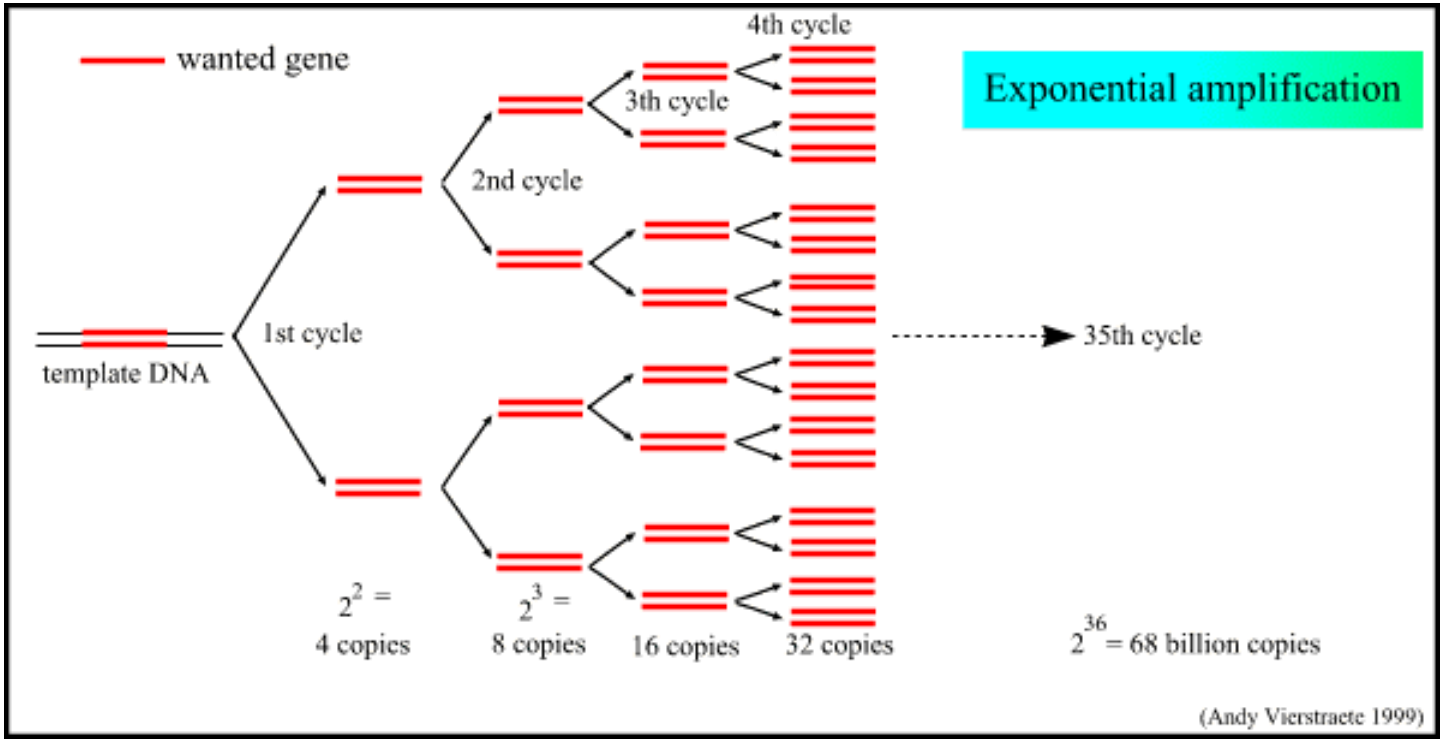
1993 Nobel Prize in Chemistry

PCR: Polymerase Chain Reaction

- A method of in vitro cloning
- Allows amplification of specific DNA molecules (fragments) in vitro through cycles of enzymatic DNA synthesis
- The most popular and widely used technique in all fields of biological studies probably.
- To amplify a lot of double-stranded DNA molecules (fragments) with same (identical) size and sequence by enzymatic method and cycling condition.

- Why?

PCR


wanted gene
4th cycle
3th cycle
2nd cycle
1st cycle
template DNA
Exponential amplification
35th cycle
2^2 =
4 copies
2^3 =
8 copies
16 copies
32 copies
2^{36} = 68 billion copies
(Andy Vierstraete 1999)

PRC diagnostics

Viruses

- HIV, SARS, H5N1

Bacteria

- meningococcus, legionellosis

Analysis for resistant genes

- MRSA, VRE

Steps in PCR

1. Denaturation of ds DNA template

2. Annealing of primers

3. Extension of ds DNA molecules

Denaturation:

- Open two strands of DNA from each other
- Boil DNA at 95 degrees for 2 minutes

Annealing

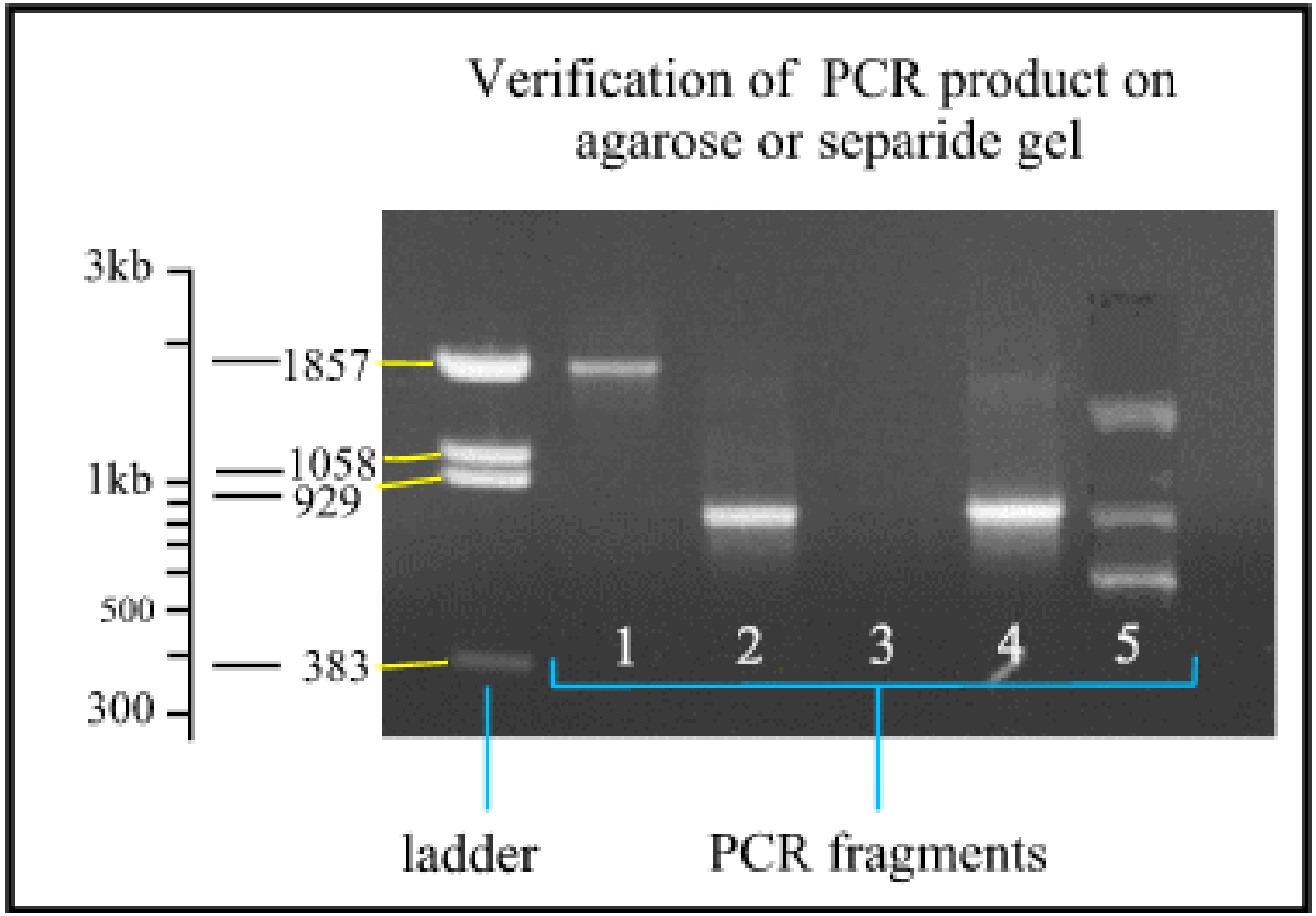
- Hybridize
- Primers anneal to denatured template DNA
  Annealing temperature 50-55 degrees

- **Extension**
- DNA polymerase synthesizes (polymerizes) new DNA molecule by adding deoxyribonucleoside complementary to the corresponding template base in a 5' to 3' direction at 68-72 degrees.

CONSTITUENTS of PCR

- Enzyme
- Buffers and $MgCl_2$
- Deoxynucleoside triphosphates (dNTPs
- Template DNA
- Primers

PCR Analysis on Agarose Gel Electrophoresis



Instrumentation